Temi Otun

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Education

University of Alberta — Edmonton, AB
Bachelor of Science, Computing Science – Artificial Intelligence Specialization, Minor in Business — *Expected Apr. 2027*

- **Awards:** Jason Lang Scholarship, Dean's List

Experience

Data Engineering Intern — May 2026 – Aug. 2026
Royal Bank of Canada (RBC) – Amplify Program — *Vancouver, BC*

- Expanded a highly imbalanced dataset from **1.4K to 200K+ synthetic samples** using data augmentation and distribution-matching techniques, enabling supervised learning and achieving **96% AUC** on a holdout test set
- Developed and benchmarked several ML models including XGBoost, LightGBM, CatBoost, SVM, and Logistic Regression for large-scale fraud text classification, improving precision from **37% to 91%**, recall from **33% to 86%**, and F1-score by **over 150%** while achieving a **94% accuracy**
- Built an end-to-end NLP pipeline that processed **1.36M+ tokens** of unstructured text, generated embeddings, classified cases with an SVM, and routed flagged cases to an LLM for secondary analysis, reducing false positives by **30%**

Machine Learning Research Intern — Jan. 2025 – Aug. 2025
The Metabolomics Innovation Centre — *Edmonton, AB*

- Enhanced long-range temperature forecasting system, resulting in up to **47%** lower MAE with an average **24.3%** improvement across a six-year test set compared to previous approaches
- Developed multivariate time series models for precipitation forecasting, incorporating **100+** climate features and outperforming baseline models in **73%** of test years
- Assembled forecasting pipelines with advanced feature engineering and benchmarked **20+** ML models, attaining a **42%** accuracy improvement over baseline methods
- Automated SQL pipelines to ingest weather and climate data from APIs and web scrapers, generating new feature combinations to support testing and improvement of forecasting models

Machine Learning Research Assistant — Sept. 2024 – Apr. 2026
University of Alberta — *Edmonton, AB*

- Contributed to **3** machine learning systems in computational psychiatry and predictive healthcare, including dementia detection and ECG signal modeling on large-scale clinical datasets
- Synthesized insights from **40+** research seminars on survival analysis and disease prediction, applying advanced statistical and ML methods to strengthen ongoing projects
- Documented and analyzed **15+** ML experiments, applying feature engineering, hyperparameter tuning, and evaluation pipelines to improve performance across classification and regression tasks

Data Management Intern — Jan. 2024 – May 2024
InfoStrux — *Vancouver, BC*

- Optimized **25+** SQL queries in Snowflake, reducing execution time by up to **60%** and boosting performance of business intelligence dashboards
- Partnered with a senior data engineer to design Snowflake staging and curated layers for **3** datasets; wrote and tuned **30+** queries, improving data reusability and cutting time-to-insight by **40%**

Projects

Emotion Detection CNN | *PyTorch, ML, Computer Vision, AI, Data Processing, Python* — GitHub

- Trained a deep convolutional neural network in PyTorch on the FER-2013 dataset with **32K+** labeled images for multi-class facial emotion recognition
- Achieved **70%** test accuracy across **7** emotion classes and integrated webcam inference for real-time emotion detection
- Applied augmentation, batch normalization, and ReLU activations to improve model performance

Lung Cancer Detection | *AI, Python, Machine Learning, AI engineering* — GitHub

- Achieved **99%** recall, **94%** accuracy, **95%** precision, and **97%** F1 score on the best classification model
- Built and compared multiple ML models, including SVM, k-NN, and Random Forest, to determine the most effective classification approach

Technical Skills

Programming: C, Java, Python, SQL
Libraries & Frameworks: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, React
Tools & Platforms: Git, Linux, Bitbucket, Docker, Snowflake, Kubernetes, AWS S3